

EXPERIENCE LAB INTERNSHIP PROGRAMME

Week 8: Final Project Summary

By Osuji Onyinyechi

HealthConnect Clinic Experience Lab*Data Science Track - Final Integration, Presentation & Project Showcase*

Programme	AnalystLab Africa Experience Lab
Week	Week 8 (Final)
Project	HealthConnect Clinic Experience Lab
Track	Data Science

1. Week 7 to Week 8 Transition

- Main Week 7 output: systematic testing of the Week 6 candidate model, going beyond aggregate metrics into segment-level error analysis.
- Most important testing result: the model's overall 66% recall was concentrated almost entirely in long-lead-time appointments (91% recall), with short-lead-time appointments barely predicted at all (6% recall).
- Major issue discovered: aggregate metrics were actively hiding this imbalance; a validated interaction feature, informed by a Data Analytics finding, was tested as a fix and did not resolve it.
- What was refined: the model's documentation and scope were sharpened to explicitly state where it is and is not suitable for use, rather than pursuing further unproven feature changes.
- Component now considered ready: the final Logistic Regression model, clearly scoped to long-lead-time appointments, with all limitations documented.
- What still needs integration: a confirmed, actioned handoff to the ML Engineering track, which was prepared but not completed within the project timeline.
- Tracks involved in final integration: Data Analytics (completed, validated) and ML Engineering (prepared, not actioned).
- What this summary demonstrates for the final presentation: how testing and honest reporting of a negative result led to a properly scoped, trustworthy final model rather than an overstated one.

2. Final Model Outcome

The final candidate model is a Logistic Regression classifier predicting patient no-shows, achieving 66% recall, 62% precision, and 0.677 ROC-AUC overall. Critically, its performance is not uniform: it correctly identifies 91% of no-shows for appointments booked 31 or more days in advance, but only 6% for appointments booked within a week. This was the single most important finding of the entire project, and it was only discoverable through segment-level testing rather than aggregate evaluation.

3. Cross-Track Collaboration Across the Project

The Data Analytics track was the primary and most productive collaboration throughout Weeks 5 through 8. Two separate validated findings were exchanged: a booking-lead-time breakdown (Week 6) and a combined booking-lead-time-and-reminder-status breakdown (Week 7). Both were independently recalculated in this project's notebooks and matched exactly, providing genuine mutual validation rather than a one-way information exchange.

A planned collaboration with the ML Engineering track did not result in an actioned exchange. Model requirements were documented and made available, but no confirmed two-way integration occurred. This is reported plainly rather than reframed as complete.

4. Key Decisions Made Across the Project

- Excluded Cancelled appointments from the modelling target, treating no-shows and cancellations as distinct behaviors.
- Selected Logistic Regression over Random Forest for interpretability, despite Random Forest's marginally better accuracy.
- Refined booking lead time bins using a cross-track-validated finding rather than an internally assumed grouping.
- Tested and rejected an interaction feature after evidence showed no improvement, rather than adopting it on the strength of the underlying finding alone.
- Chose to document the model's segment limitation honestly rather than reporting only the more favorable aggregate metrics.

5. Business Value and Contribution to HealthConnect

This work gives HealthConnect a model that is genuinely usable today for one specific, well-defined purpose: prioritizing reminder outreach for patients with appointments booked well in advance. It also gives HealthConnect an honest account of what the model cannot yet do, preventing false confidence in short-notice risk coverage, which is arguably the more operationally costly case for the clinic to manage.

6. Remaining Limitations

- Severe recall imbalance across booking lead time segments remains unresolved.
- No hyperparameter tuning or cross-validation was performed at any stage of the project.
- The dataset is synthetic; real-world generalization is unverified.
- The ML Engineering handoff remains a prepared but unconfirmed dependency.

7. Reflection on Lessons Learned

The most important lesson from this project was that a single overall performance metric can conceal a serious, business-relevant weakness. Segment-level testing, not just aggregate evaluation, was what surfaced the model's real limitation. Equally important was the discipline of reporting a validated negative result (the interaction feature test) honestly, rather than treating a small aggregate uptick as a meaningful win. Cross-track collaboration was most valuable when it produced an independently verifiable, two-way exchange, as it did with the Data Analytics track, rather than a one-sided handoff.

8. Project Closure

This concludes the Data Science track's contribution to the HealthConnect Experience Lab, spanning Week 4's problem definition through Week 8's final model documentation and presentation. The final model, its validated limitations, and the full cross-track collaboration record are documented in the accompanying notebook and the project's GitHub repository.